ET4382-SCPC Basic testbench 2025

Notes

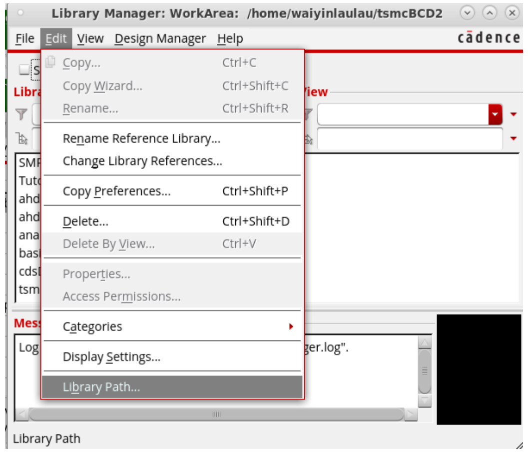
- This testbench is created according to SCPC HW1 (version: 24.03.2022) circuitA, as starting point for student to start SCPC HW.
- The circuit schematic and simulation expression are provided.
  To run this simulation, student should read through the testbench and HW1 document, calculate necessary parameter and set up simulator.
- This testbench is only a reference for student to study, not exactly a grading guideline.

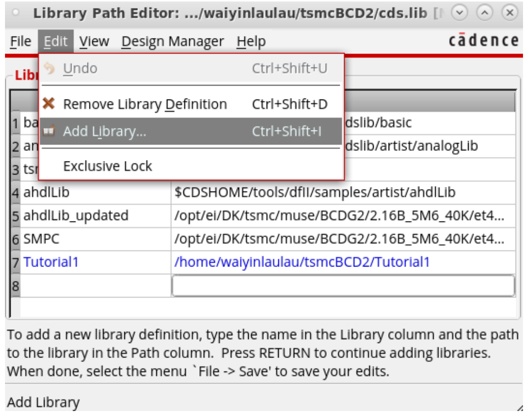
Steps set the testbench to your local(1)

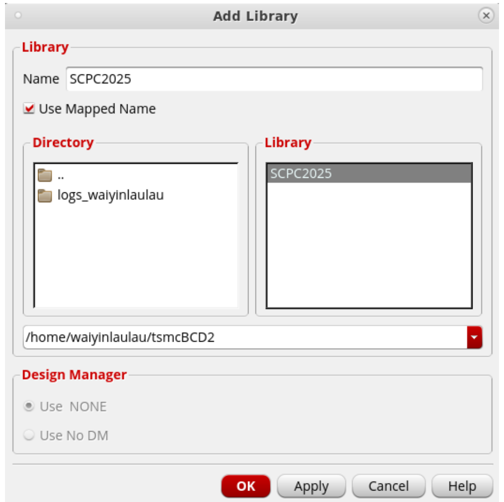
1. Copy the file from course server to your local with following command :
   “cp –r /opt/ei/DK/tsmc/muse/BCDG2/2.16B_5M6_40K/et4382/SCPC2025 ~/tsmcBCD”
   - “~/tsmcBCD” is the local folder which you created for your cadence files in tutorial1.
   - “SCPC2025” is the library created, and it contains the testbench.

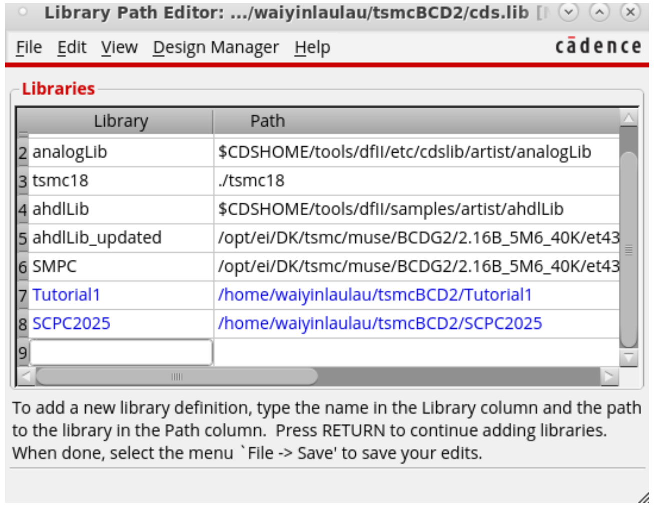
Steps set the testbench to your local(2)

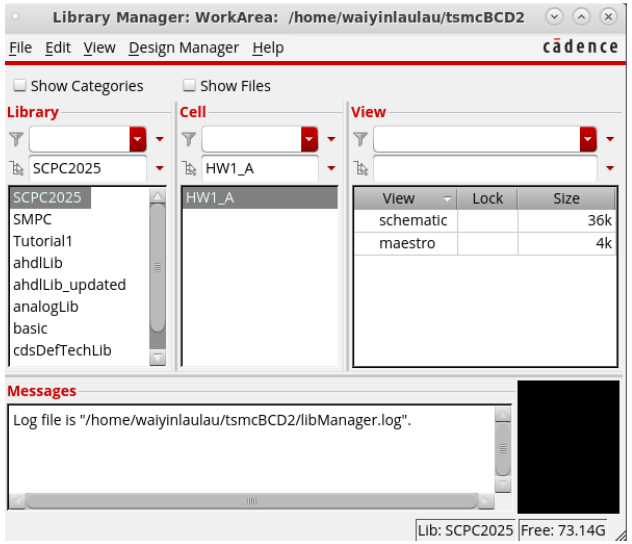
2. Add the library to your cadence



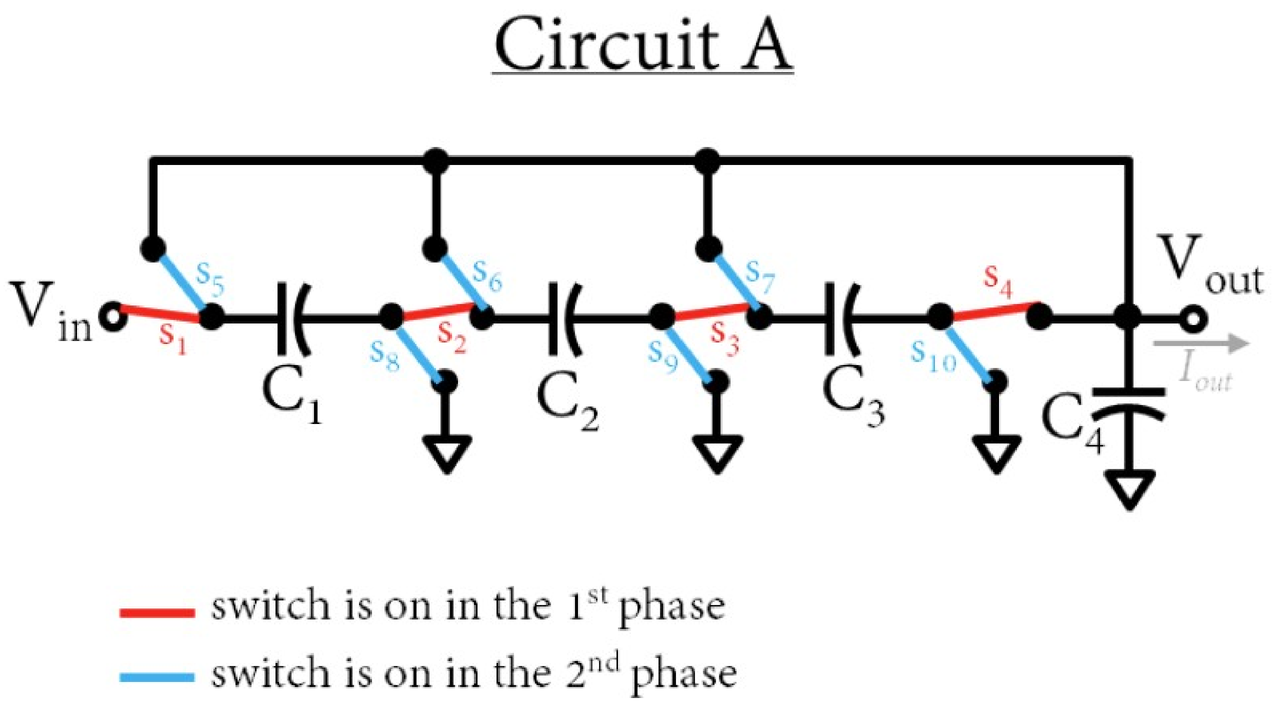




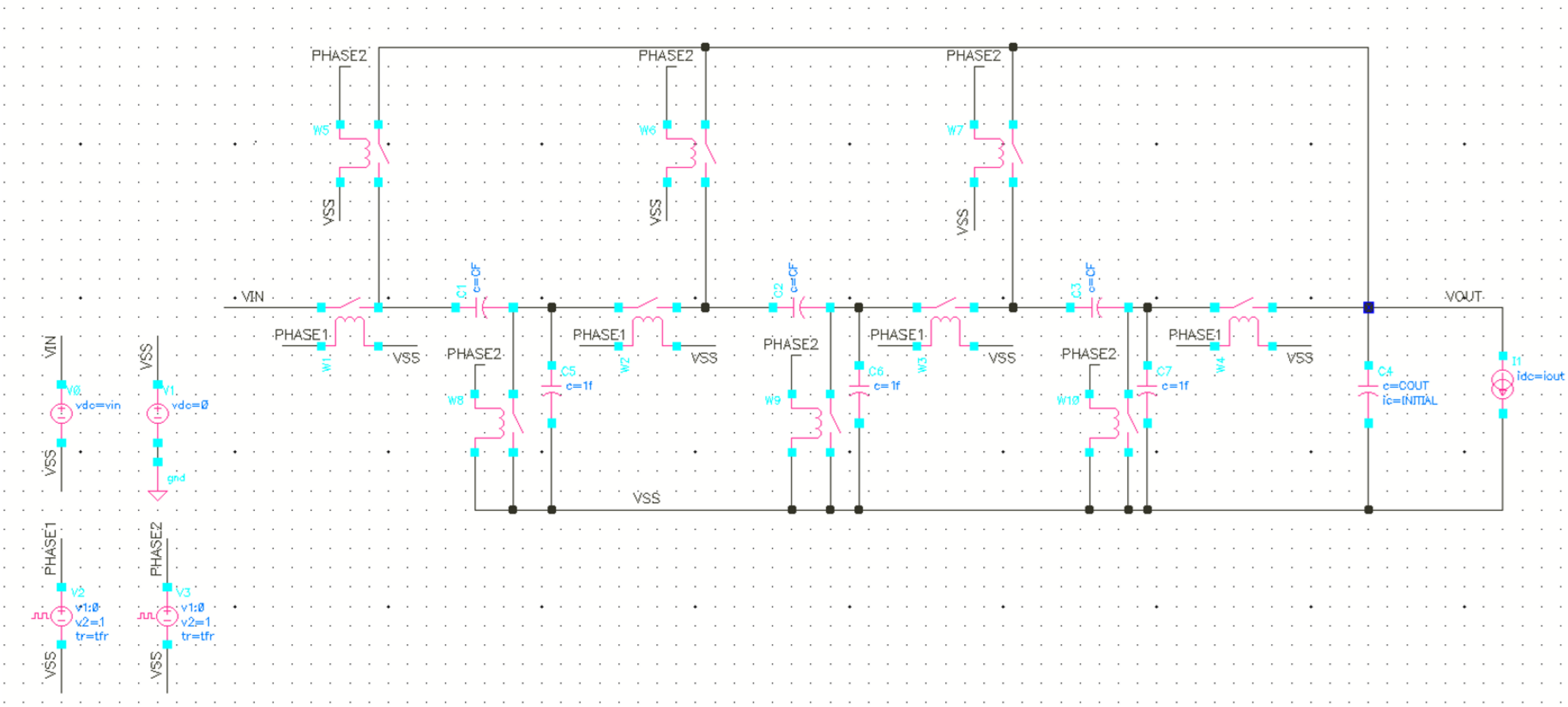




SCPC HW1:CircuitA(version: 24.03.2022)

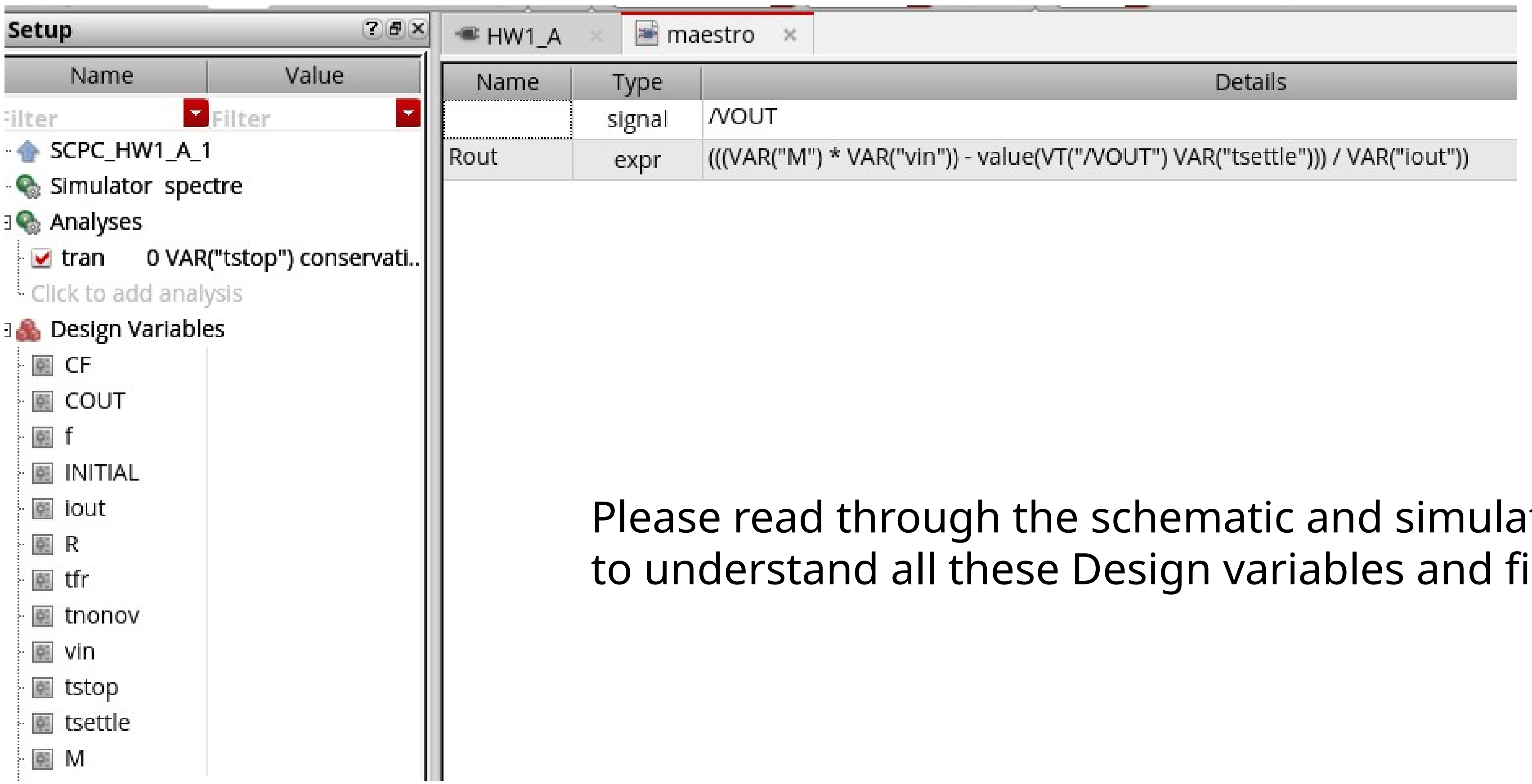


Schematic



PHASE2
PHASE2
PHASE2
W5
W6
W7
VSS
VSS
VSS
VIN
VOUT
C1
c=CF
C2
c=CF
C3
c=CF
PHASE1
PHASE1
PHASE1
PHASE1
W1
W2
W3
W4
VSS
VSS
VSS
VSS
PHASE2
PHASE2
PHASE2
W8
W9
W10
C5
c=1f
C6
c=1f
C7
c=1f
C4
c=COUT
ic=INITIAL
I1
idc=iout
VSS
VIN
V0
vdc=vin
VSS
VSS
V1
vdc=0
gnd
PHASE1
V2
v1:0
v2=1
tr=tfr
VSS
PHASE2
V3
v1:0
v2=1
tr=tfr
VSS

Simulation Setting



Please read through the schematic and simulation setting careful to understand all these Design variables and fill them in